Design and Implementation of Chinese Chess Based on Manipulator

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***Abstract*—With the development of artificial intelligence, artificial intelligence + robots have slowly entered people's field of vision. In this paper, we apply robots to Chinese chess. The design is based on the microcomputer as the control center, and through the serial communication transmission to the robot arm, by the robot arm to complete the robot chess; In terms of visual recognition, we use hough transform and character recognition to classify and identify chess pieces; In terms of chess strategies, this paper adopts alpha-beta search engine algorithm to conduct simulation experiments. The 6-layer search algorithm can complete the calculation within 5S~10S, the robotic arm movement of each step can be completed within 10S~12S, and the identification of chess pieces by taking photos can be completed within 2S.**

Keywords—*Chinese chess game robot; hough transform; alpha-beta algorithm;*

I. INTRODUCTION

With the improvement of the quality of life and the rapid development of artificial intelligence, people are more and more in contact with robots, and robots are also used in various fields, especially for the development of service and entertainment robots. Chinese chess has a long history, because its rich puzzle and entertainment attributes are very popular in China, especially for the elderly and children. However, the current aging society in China and the increase in only children, the lack of companionship between the elderly and children, so designing a chess robot that can play against people is very valuable and meaningful. In recent years, more and more experts have carried out the research work on the robot chess game and made some achievements.

As early as 1958, IBM developed the IBM704 computer that could play chess with humans.From IBM's "Deep Thought" series in 1988 to the "Deep Blue" series in 1991, the games were played by computers, with no physical robot structure[1].By the 21st century, the Ukrainian state Marine university had produced one Chess Robot, a large robotic system with no visual system.China's research on robots is relatively late, but it is developing rapidly, and many companies, universities and research institutes have also begun to study chess-playing robots.Shougang group exhibited a robot system with chess function in 2003. This robot only added two huge robot hands on the basis of welding robot, which was the transformation of industrial robot.In 2008, a chess-playing robot developed by the zhengzhou institute of machinery in China integrated all the functions of an intelligent robot and also had a voice function.With the advent of AlphaGo, machine learning and deep learning have become a hot research field. Subsequently, GAN has been applied to Chinese chess.

II. INTEGRATED SYSTEM DESIGN

The overall design of this paper is a Chinese chess robot based on machine vision. Its system structure is mainly composed of mechanical arm, robot control system, robot vision system, referee end and chessboard.The specific implementation process of chess is as follows: firstly, the camera is connected to the referee end, the mechanical arm is connected to the PC end of the player, and the PC end of the red and black sides is connected to the referee end. The referee end converts the collected checkerboard images into a one-dimensional array and sends them to the red and black sides through serial communication.Then, the chess robot will calculate the position of the next chess piece according to the chess spectrum algorithm, and send instructions to the robot arm, which will confirm the position by pushing the button after the robot arm completes the chess.Finally, the player's PC terminal will feed back the finished information to the referee terminal, waiting for the next operation instruction. [2].

A. Mechanical arm

The mechanical structure of the robot determines the trajectory of the chess robot, which is the realization of the movement function of the robot.The four-axis robot arm is assembled in this paper, and its physical picture is shown in Figure 1.

Fig. 1. Assembled four-axis mechanical arm physical map

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As can be seen from figure 1, the main joint structure of the assembled four-axis robotic arm adopted in this paper is composed of a rotary base, a big arm, a small arm and a suction cup. There are four degrees of freedom of the joints, which are similar to the waist, arm and wrist parts. The horizontal rotation of the base of the robotic arm (θ1) achieves the overall rotation of the robotic arm.The upper joint of the upper arm is used to support the lower arm, which can be used for up-and-down pitching motion (θ2, θ3), while the wrist part (θ4) is fixed and pneumatic devices are used to achieve the suction of the pieces[3].The specific rotation Angle of the robot arm is shown in Table 1:

TABLE I. ROBOT ARM AXIS ROTATION ANGLE

Each axis	θ_1	θ_2	θ_3	θ_4
Angle of rotation	-90°~90°	0°~210°	-180°~150°	-90°~90°

B. *Visual system design*

The visual system in this paper is independent of the robot.Firstly, the image collection of Chinese chess spectrum was carried out. This paper adopted ordinary household camera, with the resolution of 960×720 pixels, and the diameter of the chess pieces adopted in this paper was 25mm.

(1) circle detection and positioning: In this paper, Hough transform is used to detect and locate the collected images.Hough is one of the basic methods to recognize geometric shapes from images in image processing.The basic principle of Hough transform is to transform a given curve in the original image space into a point in the parameter space by using the duality of point and line.In this way, the detection problem of a given curve in the original image is transformed into the problem of finding the peak value in the parameter space[4].That is to say, the detection of the overall characteristics into the detection of local characteristics.Such as straight lines, ellipses, circles, arcs and so on.Because Hough transform is not sensitive to noise, and can detect multiple circles at one time.

(2) color recognition: The RGB value range of the red pieces used in this paper is: R value range is 245~255, G value range is 0~10, and B value range is 0~10.The RGB value range of the black pieces is: R value range is 0~10, G value range is 0~10, B value range is 245~255.

(3) character recognition: Character recognition is template matching, but in pieces put a piece of the process of put can be anywhere, it caused a lot of trouble to template matching.In this paper we uesd a rotating difference identification algorithm[5], specific it is to identify a picture with the module of image pixel difference, then, will continue to do template rotation α angle difference, keep the value of the minimum order and the remaining 6 template for the operation, finally, the minimum is to recognize the value of the corresponding characters pieces of identification results.

III. HARDWARE DESIGN

A. *camera and rack*

In this paper, the camera with 960×720 resolution is used. The camera is connected to the referee. By adjusting the threshold of the camera, it can achieve non-reflective, not too bright and too dark. The photo can be recognized by the naked eye normally. In the process of playing chess, the camera is obtained. Chess information is sent to both red and black. The camera frame is located at the center of the chessboard directly above, in order to capture more complete board information, the structure of which is shown in Figure 2.

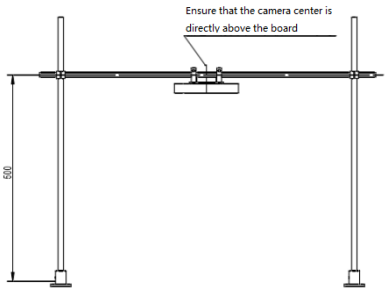

Fig. 2. Camera and frame structure

B. *Steering Gear*

In this paper we uses XQtor-H7 digital steering gear. Compared with other steering gears, XQtor-H7 digital steering gear has the advantages of high torque, high resolution and full angle. The output torque can reach 7Nm, the minimum resolution angle is 0.08°, 360 degrees. No dead angle rotation, and support RS-485 bus communication. The specific detailed parameters are show in Table 2.

Fig. 3. XQtor-H7 digital steering gear

TABLE II. XQTOR-H7 DIGITAL STEERING GEAR PARAMETERS

Rated voltage	8-14.8*V*
rated power	13*W*
Maximum speed	60*RPM*
Rated torque	1.5*NM*
Maximum torque	7*Nm*
Gear reduction ratio	200:1
Angle resolution	12 *digits*
Operating temperature	10-80 *Celsius*
Communication Interface	*RS*-485
Protocol	*Homemade agreement*
Gear back	<=0.6°
weight	146*g*

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C. NanoPi M2A Development Board

The NanoPi M2A used in this article is another open source handheld maker created by the friendly electronics team for enthusiasts such as makers, embedded enthusiasts, and electronic artists. The NanoPi M2A uses the Samsung quad-core Cortex-A9 architecture S5P4418 processor. Compared with the friendly NanoPi M2, it not only integrates the NanoPi M2 interface, but also integrates WiFi and Bluetooth. The interface and layout are completely the same as the NanoPi M3.The NanoPi M2A is equipped with a video input/output interface (DVP Camera/LVDS/HDMI/LCD interface), an I2S interface, a 3.5mm audio output interface, and four USB ports. Supports power management functions, on-board ceramic antenna with serial port debugging. In software, NanoPi M2A runs Ubuntu Core with Qt, Android4.4, Android5.1, Debian8.1 and other systems. The kernel driver source is completely open source, and it provides a wealth of sample code and libraries.[6]

IV. SOFTWARE DESIGN AND IMPLEMENTATION

A. Board Information Conversion

In this paper, Chinese chess spectrum is represented by coordinates, which are represented by the coordinates of 10 rows and 9 columns. The specific coordinates are as shown in Figure 4:

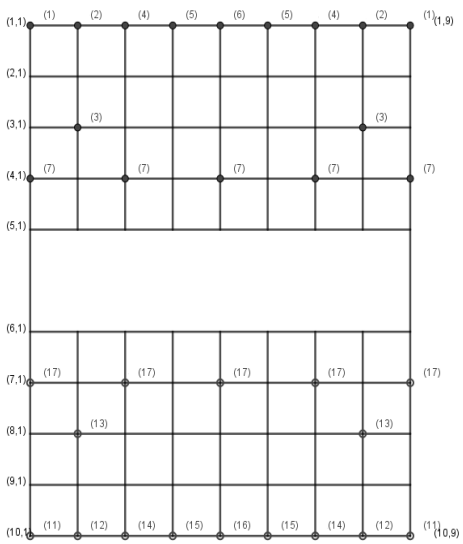

Fig. 4. Chinese chess board information coordinate representation

In this paper, the information of each piece is represented by Numbers: red chess is represented by 1-7, and black chess is represented by 11-17:

(1)Red Chess: Rooks 1, Knights 2, Cannons 3, Elephants 4, Warrior 5, King 6,Pawns 7.

(2)Black chess: Rooks 11, Knights 12, Cannons 13, Elephant 14 ,Warrior 15 ,King 16, pawn17.

(3)The position of no chess piece is represented by 0.

(4)In the initial position, the chess game information of Chinese chess is as follows:

[1, 2, 4, 5, 6, 5, 4, 2, 1,
0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 3, 0, 0, 0, 0, 0, 3, 0,
7, 0, 7, 0, 7, 0, 7, 0, 7,
0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0,
17, 0, 17, 0, 17, 0, 17, 0, 17,
0, 13, 0, 0, 0, 0, 0, 13, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0,
11, 12, 14, 15, 16, 15, 14, 12, 11]

B. Chinese chess situation evaluation function

The so-called chess game evaluation is to evaluate and quantify (score) the current chessboard situation, in order to determine the next move of chess in the search process by relying on the value of its value.The evaluation of the situation in the chess game is a very important part. The correctness of the current chess situation evaluation will directly affect the final development and trend of the entire chess game. At present, the situation assessment method commonly used by computers in Chinese chess games mainly uses static estimation methods.

According to the unique rules of Chinese chess, the evaluation function mainly includes the value of each piece, the value of the position of the piece, the control of the piece to the board, the flexibility of the piece, the threat and protection between the pieces, and some influence on the game. feature. Adding all the different scores or scores is the sum of the so-called ones:

$$S\text{va}\models\sum \mathrm{K}_i F_i(P) \tag{1}$$

Where K_i is the characteristic coefficient, F_i is the evaluation feature function, and P is the current situation of the search. Thus, the sum of the value of the red side is called R_V, and the sum of the value of the black side is called B_V. Then the difference between the sum of the value of the red side and the black side is the valuation of the whole situation. We use Value to indicate:

$$V\text{alue} = R_V - B_V \tag{2}$$

C. Alpha-Beta algorithm

Chinese chess programs use the "search" function to find the law. The search function gets the game information and then looks for the best move for the program side. Among the Chinese chess algorithms, the most commonly used search algorithms are: minimum-maximum search, alpha-beta search, and iterative deepening[7].

In this paper, we uses the Aplha-Beta algorithm, which is based on the maximal minimum search algorithm. The maximal and minimal search strategy is to consider a relatively good way to move from a possible step after considering the two steps of the game. It can be understood as specifying a limited search depth by solving the problem within a limited search range.

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To this end, we define a static estimation function F to make an estimate of the strength of the game. This function can be defined according to the characteristics of the game's strengths and weaknesses. Here, it is stipulated that MAX represents the red side, MIN represents the black side, and P represents a chess game (ie, a state). Conducive to the potential state of MAX, F(p) takes a positive value, which is beneficial to the potential state of MIN, F(p) takes a negative value, the potential state is equalized, and F(p) takes zero. The basic idea of a very small search is:

(1) When it is the turn of MIN to move the node, MAX should consider the worst case (hence, F(p) takes a minimum value).

(2) When it is the node of MAX's walking, MAX should consider the best case (hence, F(p) takes the maximum value).

(3) When the evaluation goes backwards, corresponding to the confrontation strategy of the two players, the reversed values are transmitted upwards using the two methods (1) and (2) alternately on different layers. So this search method is called a very small process. In fact, this algorithm assumes that both sides are the best step in the simulation process. For the MAX side, the best step for the MIN side is the worst case, and MAX is constantly maximizing its own interests.

Alpha-beta is optimized on this basis. In fact, MIN and MAX are connected in the process of continuous reversal. When they satisfy a certain relationship, subsequent searches are superfluous! The alpha-beta pruning algorithm combines the generation of post-regression and back-pushing estimates to reduce some useless branches in time to improve the efficiency of the algorithm.

The lower bound of the defined maximal layer is alpha, the upper bound of the very small layer is beta, and the alpha-beta pruning rules are described as follows:

(1) alpha pruning. If the beta value of any minimum value layer node is not greater than the alpha value of any of its precursor maxima layer nodes, ie alpha (precursor layer) >= beta (subsequent layer), the minimum value layer can be terminated. The search process below this MIN node. The final back-push value of this MIN node is determined as this beta value.

(2) beta pruning. If the alpha value of any maximal layer node is not less than the beta value of any of its precursor minima layer nodes, ie alpha (subsequent layer) >= beta (predecessor layer), the maxima layer can be terminated. In the search process below this MAX node, the final pushback value of this MAX node is determined as the alpha value.

V. EXPERIMENTAL RESULTS

The Chinese chess table robot arm designed in this paper can realize the Chinese chess game well. After testing, it takes about 12S to move from the robot arm to the back. For example, if you want to eat the food during the movement, the whole process needs about 15S. The machine is configured as Inter(R) Core(TM) i7-7700 CPU @ 3.6GHz, 16GB memory, using 6-layer Aplha-Beta search algorithm, which can be calculated in 5S~10S.

VI. CONCLUSION

This paper is based on the current popular chess robot, and designs a small, practical and portable artificial intelligence Chinese chess robot, which has certain practical significance and reference value for the application of robots. The main conclusions of this design are as follows:

(1) In terms of chessboard information representation, 10×9 two-dimensional coordinates are used, which simplifies the code amount of the program.

(2) In the aspect of chess recognition, the Hough transform object in opencv is used for positioning and calibration, and then the chess piece calibrated on the chessboard is compared with the rule picture that we have previously calibrated to achieve the purpose of recognition.

(3)In terms of search algorithm, this paper compares Alpha-Beta algorithm and improved algorithm, and calculates the calculated response time and final score of each algorithm, and selects Alpha-Beta historical heuristic algorithm.

The chess robot in this paper has to complete the game in the specified time, coupled with the limited hardware conditions and computer processing capabilities, so this design does not use the Q-Learning reinforcement learning algorithm, prepared in the follow-up study and life. Test the performance of the algorithm in Chinese chess.

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